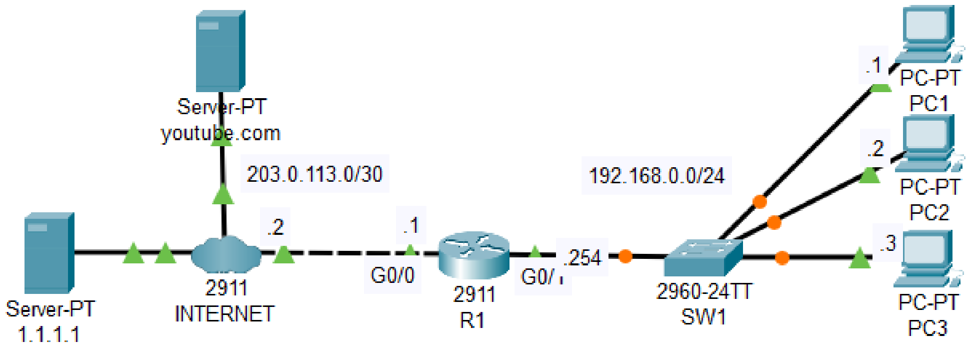
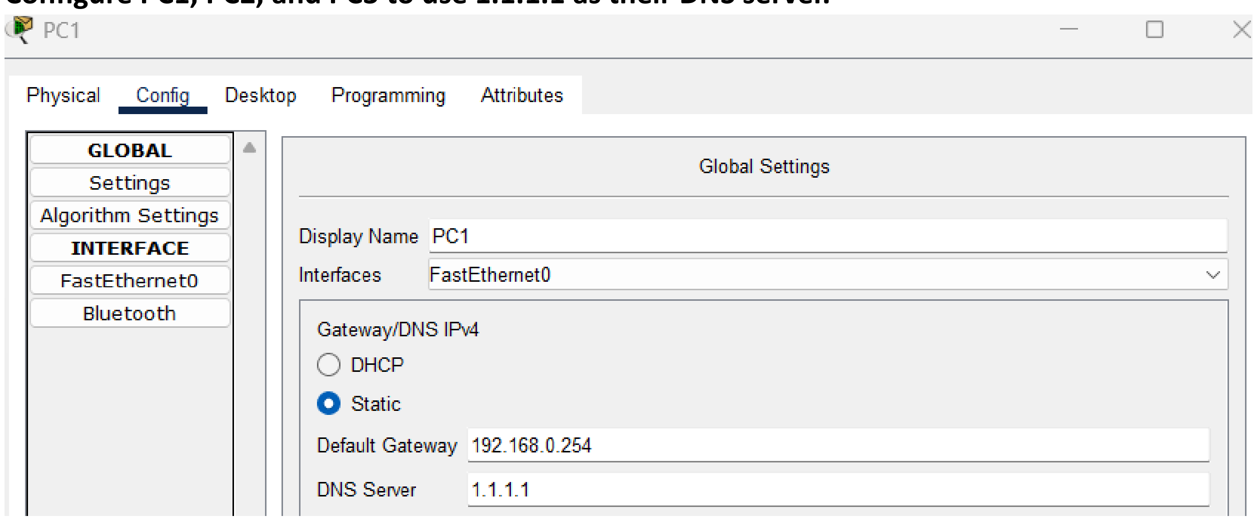


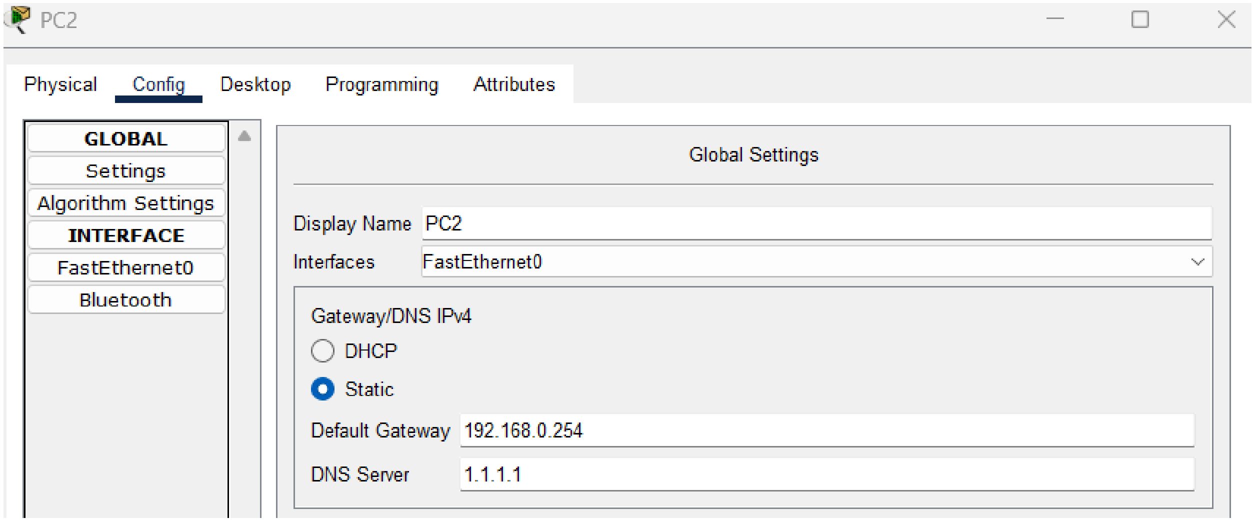
Topology

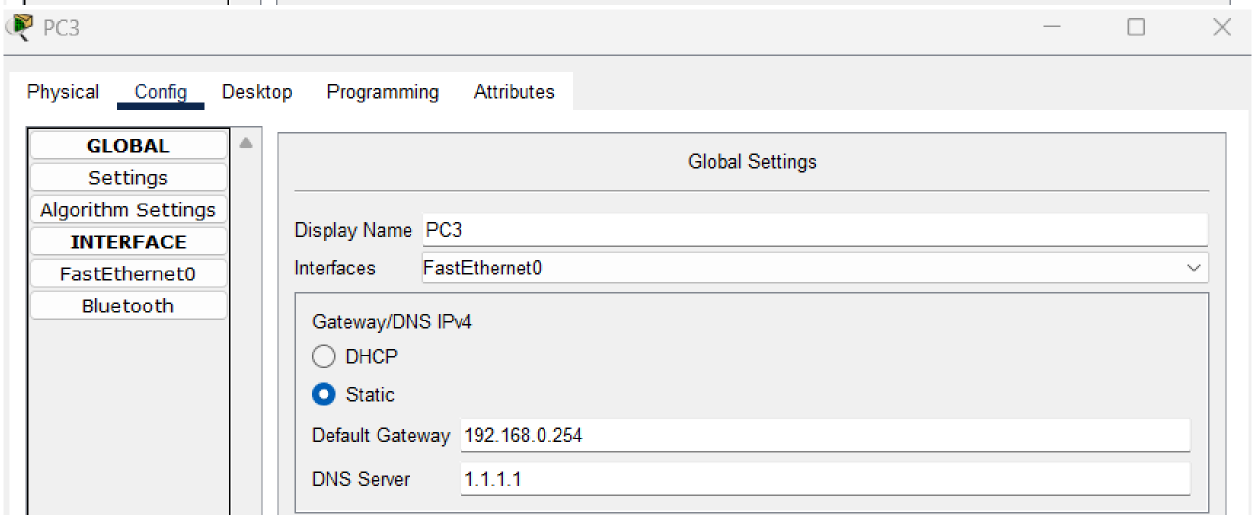
1. Configure a default route to the Internet on R1.

```
R1>en
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ip route 0.0.0.0 0.0.0.0 203.0.113.2
R1(config)#do ping 1.1.1.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 1.1.1.1, timeout is 2 seconds:
..!!!
Success rate is 60 percent (3/5), round-trip min/avg/max = 0/2/6 ms
```

2. Configure PC1, PC2, and PC3 to use 1.1.1.1 as their DNS server.





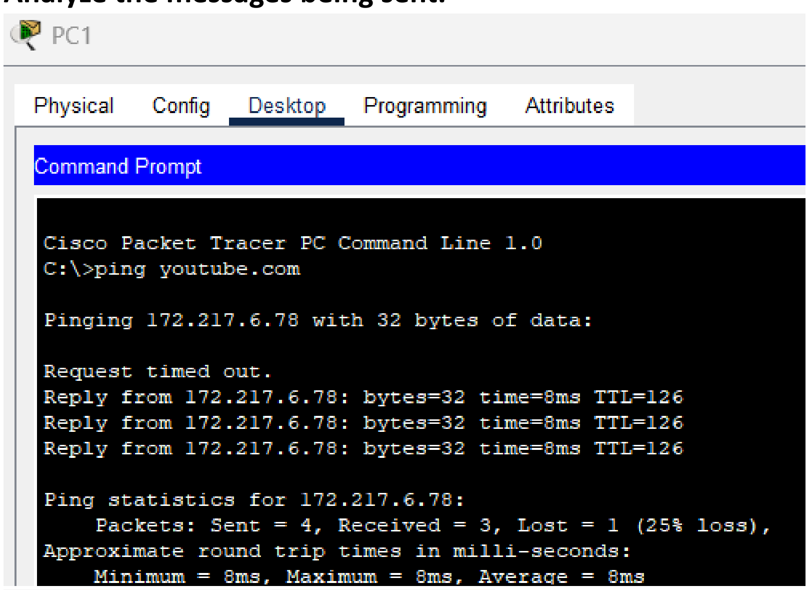


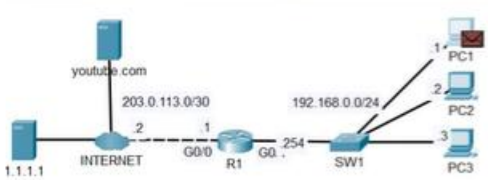
3. **Configure R1 to use 1.1.1.1 as its DNS server. Configure host entries on R1 for R1, PC1, PC2, and PC3. Ping PC1 by name from R1.**

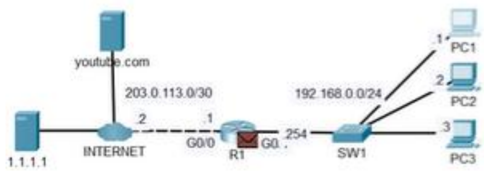
```
R1(config)#ip name-server 1.1.1.1
R1(config)#ip host R1 192.168.0.254
R1(config)#ip host PC1 192.168.0.1
R1(config)#ip host PC2 192.168.0.2
R1(config)#ip host PC3 192.168.0.3
R1(config)#do sh hosts
Default Domain is not set
Name/address lookup uses domain service
Name servers are 1.1.1.1

Codes: UN - unknown, EX - expired, OK - OK, ?? - revalidate
       temp - temporary, perm - permanent
       NA - Not Applicable None - Not defined

Host                      Port  Flags      Age Type   Address(es)
PC1                       None  (perm, OK)  0   IP      192.168.0.1
PC2                       None  (perm, OK)  0   IP      192.168.0.2
PC3                       None  (perm, OK)  0   IP      192.168.0.3
R1                        None  (perm, OK)  0   IP      192.168.0.254
R1(config)#do ping PC1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.0.1, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms
```

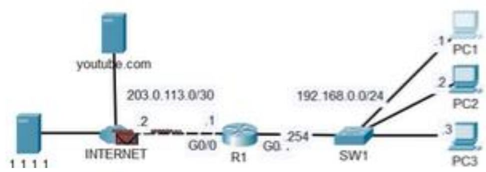
R1 has an entry for PC1, so it is able to resolve the hostname PC1 to its IP address. The IP domain lookup command must be enabled for R1 to do this, but it is usually enabled by default.

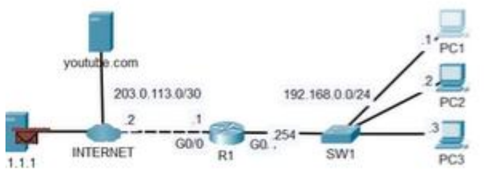
4. **#USE SIMULATION MODE FOR THIS STEP#. From PC1, ping youtube.com by name. Analyze the messages being sent.**

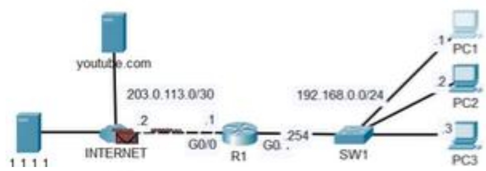


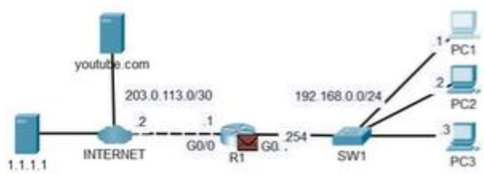


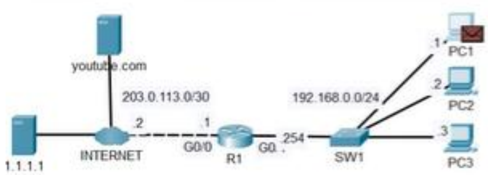












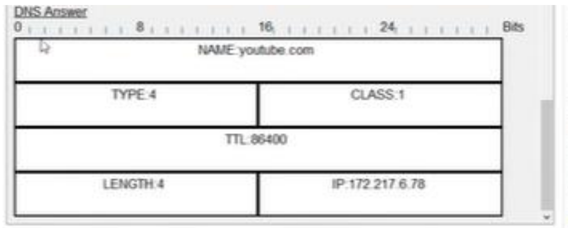
Simulation Panel

Event List

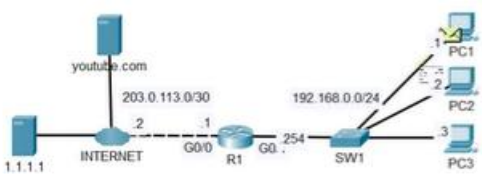
Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	PC1	DNS
	0.001	PC1	SW1	DNS
	0.002	SW1	R1	DNS
	0.003	R1	INTER...	DNS
	0.004	INTERNET	1.1.1.1	DNS
	0.005	1.1.1.1	INTER...	DNS
	0.006	INTERNET	R1	DNS
	0.007	R1	SW1	DNS
	0.008	SW1	PC1	DNS
	0.008	--	PC1	ICMP

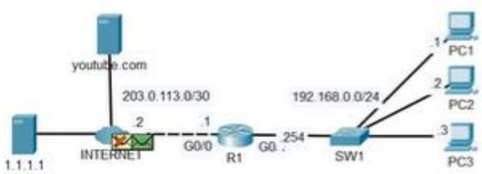
Because R1 already pinged PC1, PC1 knows the MAC address of its default gateway R1, so PC1 does not need to send an ARP request. The DNS messages goes to R1, to the Internet, to 1.1.1.1, before being sent back.

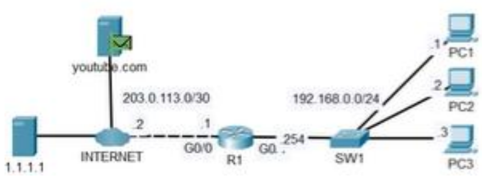
After 1.1.1.1's DNS response is back at PC1, here are the contents.

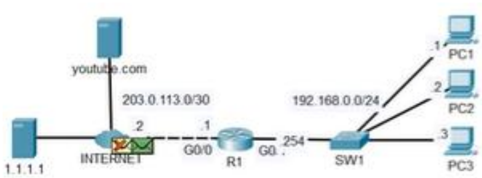


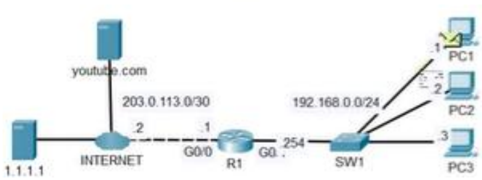
youtube.com’s IP address is shown: 172.217.6.78.

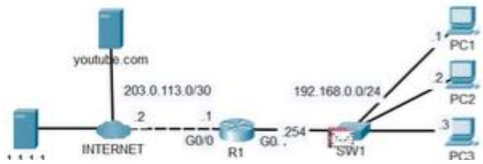


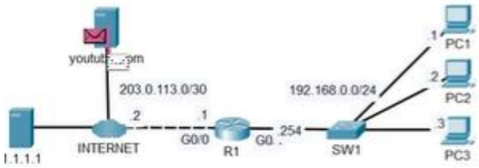


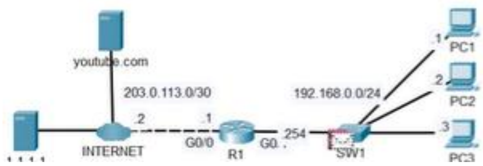












Simulation Panel

Event List

Vis.	Time(sec)	Last Device	At Device	Type
	0.008	--	PC1	ICMP
	0.009	PC1	SW1	ICMP
	0.010	SW1	R1	ICMP
	0.011	R1	INTER...	ICMP
	0.011	--	INTER...	ARP
	0.012	INTERNET	youtub...	ARP
	0.013	youtube....	INTER...	ARP
	1.447	--	SW1	STP
	1.448	SW1	PC1	STP
	1.448	SW1	PC2	STP
	1.448	SW1	PC3	STP
	1.448	SW1	R1	STP
	3.446	--	SW1	STP

Simulation Panel

Event List

Vis.	Time(sec)	Last Device	At Device	Type
	5.446	SW1	PC1	STP
	5.446	SW1	PC2	STP
	5.446	SW1	PC3	STP
	5.446	SW1	R1	STP
	6.012	--	PC1	ICMP
	6.013	PC1	SW1	ICMP
	6.014	SW1	R1	ICMP
	6.015	R1	INTER...	ICMP
	6.016	INTERNET	youtub...	ICMP
	6.017	youtube....	INTER...	ICMP
	6.018	INTERNET	R1	ICMP
	6.019	R1	SW1	ICMP
	6.020	SW1	PC1	ICMP

The message goes to R1, and the Internet, but now the Internet router has to send an ARP message to Youtube.com, because traffic has not been sent yet between them During that time, PC1's first ping fails, but then it sends another ping and it reaches youtube.com, and youtube.com replies.